Chapter 10

HDR Setup and Grading

High Dynamic Range (HDR) grading for cinema, television, and streaming is the latest evolution of the consumer media experience.While HDR workflows in high-end cinema and television aren't new, this way of mastering media has been slow to expand to less expensive programming.

However, new developments and an expanding array of affordable HDR-capable consumer devices are poised to make HDR mastering of visual content increasingly ubiquitous.

This chapter describes what HDR is for the uninitiated and covers the operational details that will let you set up DaVinci Resolve to do HDR grading.

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High Dynamic Range (HDR) Grading in DaVinci Resolve

The HDR features found in DaVinci Resolve are only available in DaVinci Resolve Studio.

High Dynamic Range (HDR) video describes an emerging family of video encoding and distribution technologies designed to enable a new generation of television displays to play video capable of intensely bright highlights and increased saturation. The general idea is that the majority of an HDR image will be graded similarly to how a Standard Dynamic Range (SDR) image is graded now, with the shadows and midtones being mostly the same between traditionally SDR and HDR-graded images. This is mostly because shadows are shadows and are meant to be dark; however this philosophy also maintains a comfortable viewing experience and easier backward compatibility when you need to master both SDR and HDR versions of a program. The difference is that HDR provides abundant additional headroom for very bright highlights and color saturation that far exceed what has been previously visible in SDR television and cinema. This enables the colorist to create more vivid and life-like highlights in images, such as sunsets, lit clouds, firelight, explosions, sparkles, and other intensely bright and colorful imagery. In short, you can now “open up” the highlights in an image just as you've always been able to open up, or expand, the detail of the shadows. This not only provides more life-like lighting intensity and saturation, but it also dramatically expands the contrast available in the scene. For example, a calibrated SDR display should have a peak luminance level of 100 nits (cd/m^2), but existing HDR displays can provide peak luminance levels of 700, 1000, or even 4000 nits.

However, because it's an evolving technology, the technical standards that have been developed far exceed what current consumer televisions, projectors, phones, and tablets are capable of. At the time of this writing, consumer televisions are capable of outputting 700 to 1600 nits. Furthermore, consumer displays are often saddled with automatic brightness limiting (ABL) circuits that limit power consumption to acceptable levels for home use, which means that only a certain percentage of the picture may reach these peak values at any one time. This is fine, because the point of HDR is not that you're making the entire image brighter, it's that you have more headroom for specific bright highlights and additional saturation.

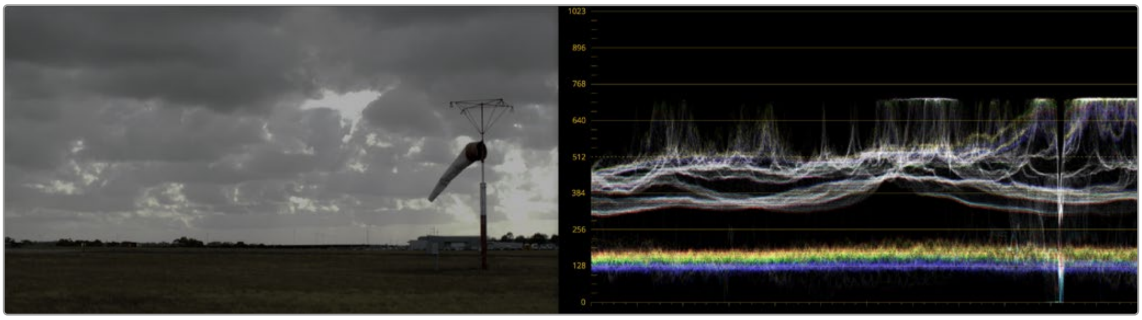
For all of these reasons, HDR standards focus on describing what displays should be capable of, not how these levels are to be used. That is a creative decision.

HDR Isn't Just for Televisions

Lest you think that living room televisions and projectors are the only way to watch HDR content, certain flagship iOS and Android phones and tablets have implemented HDR viewing capabilities that are capable of meeting or even exceeding the UltraHD requirements for HDR content on an OLED display. This makes HDR, surprisingly, a widely available mobile experience.

The Different Ways of Mastering HDR

While different HDR technologies use different methods to map the video levels of your program to an HDR display's capabilities, they all output a "near-logarithmically" encoded signal that requires a compatible television that's capable of correctly stretching this signal into its "normalized" form for viewing. This means if you look at an HDR signal that's output from the video interface of your grading workstation on an SDR display, it will look flat, desaturated, and unappealing until it's plugged into your HDR display of choice.



A graded HDR image being output looks similar to a log-encoded image

At the time of this writing, there are five principal approaches to mastering HDR that DaVinci Resolve is capable of supporting, including:

- Dolby Vision®
- HDR10
- HDR10+
- HDR Vivid
- Hybrid Log-Gamma (HLG)

Each of these HDR standards define how an HDR signal is encoded for export and later mapped to the visible output of an HDR or SDR display. Grading to each of these standards requires some degree of color management, and DaVinci Resolve gives you three main ways to handle this:

- The easiest way is to enable Resolve Color Management (RCM) or ACES in the Color Management panel of the Project Settings, and use the Color Space conversion options that are available. There are options there for each supported type of HDR.
- The transforms that are available in RCM are also available as Resolve FX operations, if you want to organize your grading pipeline more manually using the Color Transform Resolve FX adjustment.
- LUTs are also available to accomplish each of these color space conversions if you want to develop your own specific image processing pipeline based on custom-made LUT or DCTL transforms.

Overall, Resolve Color Management and ACES are reliable and recommended approaches to handling HDR grading in DaVinci Resolve in most instances.

For more information about Resolve Color Management, *see Chapter 9, "Data Levels, Color Management, and ACES."*

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What Do I Do With HDR?

While these standards make HDR mastering and distribution possible, they have nothing to say about how these HDR-strength levels should be used creatively. That's up to you, because the question of how to utilize the expansive headroom for brightness and saturation that HDR enables is fully within the domain of the colorist, as a series of creative decisions that must be made regarding how to assign the range of highlights that are available in your source media to the above-100 nit HDR levels you're mastering to as you grade, given the peak luminance level that you're assigned to master with. Which HDR peak luminance level you use (1000 nit, 3000 nit, 4000 nit) probably depends on which display you have access to and who's distributing the resulting program.

Analyzing HDR Signals Using Video Scopes

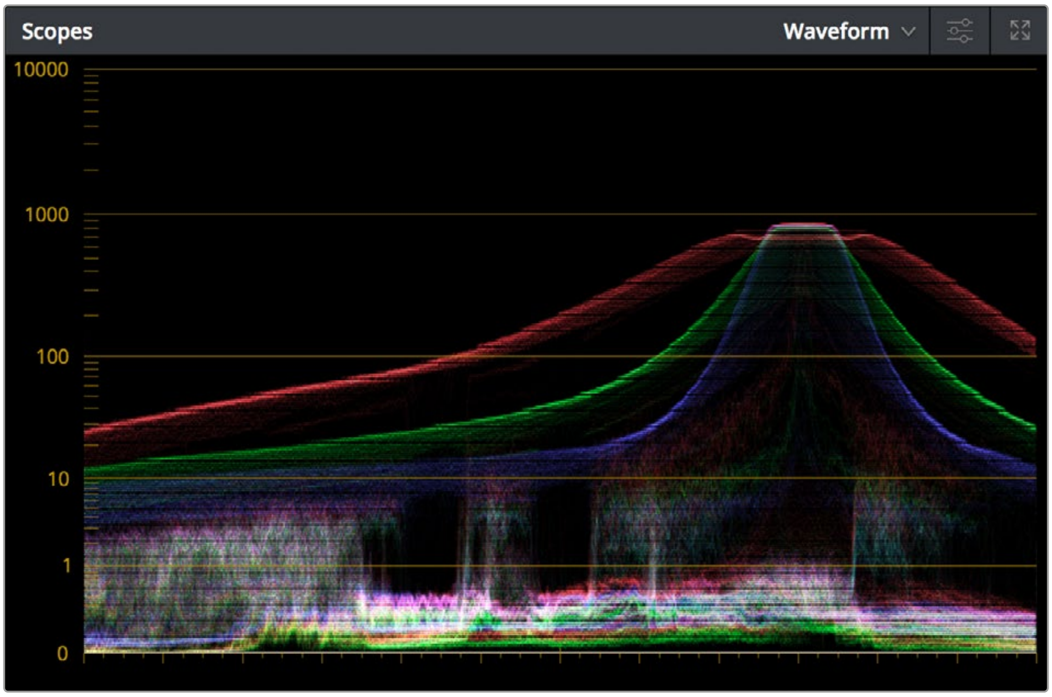
When you're using waveform scopes of any kind, including parade and overlay scopes, the signal will fit within the 10-bit scale used to analyze the signal much differently owing to the way HDR is encoded. The following chart of values will make it easier to understand how each level in "nits" (i.e., cd/m^2) corresponds to a code value within the 10-bit image scale:

10-Bit Code	**Nearest Value in cd/m^2**	**HDR Display Peak Luminance Capability**
1019[*]	10,000	No commercially available display
948	5000	Outdoor LED Displays
920	4000	Professional HDR Displays from Sony, Dolby, Flanders Scientific, EIZO, etc.
889	3000	Professional HDR Displays from Sony, Dolby, Flanders Scientific, EIZO, etc.
844	2000	Professional HDR Displays from Sony, Dolby, Flanders Scientific, EIZO, etc.
824	1600	Virtual Production Wall Panels
767	1000	Professional HDR Displays from Sony, Dolby, Flanders Scientific, EIZO, etc.
728	700	OLED mobile phone brightness
691	500	Minimum standard for an "UltraHD" OLED display

10-Bit Code	Nearest Value in cd/m²	HDR Display Peak Luminance Capability
635	300	Professional SDR displays in “HDR preview mode”
593	203	BT.2408 recommendation for diffuse white of SDR content being intercut with 1000 nit max HDR content
528	108	Dolby Cinema projector
520	100	Standard peak luminance for SDR displays
447	48	Standard peak luminance for SDR DCI projection, Dolby Cinema 3D peak luminance
4†	0	Absolute black

† 0–3 and 1020–1023 are reserved values

While this table of values is useful for understanding where HDR nit levels fall on legacy external scopes, if you’re monitoring with the built-in video scopes in DaVinci Resolve, you can turn on “HDR (ST.2084/HLG)” in the Waveform Scale Style settings in the Scopes option menu, which replaces the 10-bit scale of the video scopes with a scale based on nit values (or cd/m²) instead.



The video scopes with HDR (ST.2084/HLG) on in the Waveform Scale Style settings in the Scopes option menu

TIP: If you’re unsatisfied with the amount of detail you’re seeing in the 0–519 range (0–100 nits) of the video scope graphs, then you can use the 3D Scopes Lookup Table setting in the Color Management panel of the Project Settings to assign the appropriate “HDR X nits to Gamma 2.4 LUT,” with X being the peak nit level of the HDR display you’re using. This converts the way the scopes are drawn so that the 0–100 nit range of the signal takes up the entire range of the scopes, from 0 through 1023. This will push the HDR-strength highlights up past the top of the visible area of the scopes, making them invisible, but it will make it easier to see detail in the midtones of the image.

HDR Viewers

You can display the native viewers in DaVinci Resolve in HDR on Mac and Windows systems. The setting affects all DaVinci Resolve viewers, including the main page viewers, Cinema Mode, Fairlight Floating Viewer, Scene Cut Dialog, and the Video Clean Feed.

You will need an HDR-capable display and to turn on HDR in your OS:

- Activate the Windows HDR display settings (System > Display > HDR).
- Activate the Mac HDR display settings (System Settings > Displays > High Dynamic Range).
- In DaVinci Resolve Preferences > System > General, you will need to check the “Use 10-bit precision in viewers if available” box.

NOTE: The general controls/UI and non-viewer dialogs (e.g., secondary screen, project manager, floating Media Pool) are not affected.

Dolby Vision®

Long a pioneer and champion of HDR for enhancing the consumer video experience, Dolby Laboratories has developed a method for mastering and delivering HDR called Dolby Vision. As with most HDR standards discussed in this chapter, Dolby Vision uses the PQ (perceptual quantizer) electrical-optical transfer function (EOTF, which defines how an electronic video signal is presented on a display), which is defined by SMPTE ST.2084, along with a hierarchy of metadata that’s embedded alongside the video stream. All metadata used by Dolby Vision is organized into levels, of which the following are important to the colorist:

- Level 0 metadata, which is global metadata that defines the Mastering Display (what the colorist is using), including aspect ratio, frame rate, color encoding and information on all the target displays that are used for the Level 2 and Level 8 trim metadata below.
- Level 1 metadata, which is the Dolby Vision v2.9 analysis metadata that’s generated automatically when you use the Dolby Vision controls to analyze the clips in the timeline. The controls for automatically generating Level 1 metadata are available to all DaVinci Resolve Studio users.
- Level 2 metadata, which is the Dolby Vision v2.9 trimming metadata that’s set by the colorist via the version 2.9 trim controls available in the Dolby Vision palette of the Color page. This trimming allows adjustment of how the Dolby Vision image is to be mapped to a target display (such as a 100 nit BT.709 display) that’s different from the mastering display (such as a 1000 nit BT.2020 display). The purpose of this metadata is to maintain a program’s artistic intent by providing guidance from the colorist over how the program’s signal should be fit into the differing luminance ranges of a variety of displays with different peak luminance capabilities. Manually adjustable Level 2 metadata is only available to DaVinci Resolve Studio users via a license obtained from Dolby.
- Level 3 metadata, which is the offset for Dolby Vision v4.0 added to Level 1 metadata generated by the analyze buttons in the Dolby Vision controls. It also stores the mid tone offset data.
- Level 5 metadata, which provides information about the aspect ratio of the deliverable format, and the aspect ratio of the actual image within that format. This metadata is also applicable at the per clip level.

- Level 6 metadata, which stores the MaxCLL and MaxFALL levels required by the HDR10 mastering standard of HDR.
- Level 8 metadata, which is the updated Dolby Vision v4.0 trimming metadata that's set by the colorist via the v4.0 trim controls available in the Dolby Vision palette of the Color page. This evolved set of trimming commands allows more detailed adjustment of how the Dolby Vision image is to be mapped to a target display (such as a 100 nit BT.709 display) that's different from the mastering display (such as a 1000 nit BT.2020 display). Just like Level 2 metadata, the purpose of Level 8 metadata is to maintain a program's artistic intent by providing guidance from the colorist over how the program's signal should be fit into the differing luminance ranges of a variety of displays with different peak luminance capabilities. Manually adjustable Level 8 metadata is only available to DaVinci Resolve Studio users via a license obtained from Dolby. Whether you use Level 2 trim controls or Level 8 trim controls depends on the Dolby Vision version setting you choose in the Color Management panel of the Project Settings.

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NOTE: It's currently recommended for all users to choose Dolby Vision v4.0 for analysis and trimming, as it provides superior results. If you're required to deliver Dolby Vision v2.9 metadata when mastering for backwards compatibility, DaVinci Resolve can now export v2.9 metadata from projects using v4.0 workflows.

The metadata levels described above are current of this writing. However Dolby Vision is a rapidly evolving technology, and as Dolby adds new features and metadata levels you should reference Dolby's website to keep track of the latest developments: https://professionalsupport.dolby.com/s/article/Dolby-Vision-Metadata-Levels?language=en_US

For the foreseeable future, the current consumer display landscape encompasses a wide variety of differently performing televisions and projectors that are guaranteed to improve year over year. This means that mastering for today's displays may render content less vibrant than content that emerges five years from now. This can be especially vexing for narrative content that will have a long lifespan on streaming services as new generations of viewers discover them. While one way of solving this would be to re-grade your program many times at a variety of nit levels to create deliverables suitable to a range of display capabilities, that's an enormous amount of work.

Dolby Vision offers a shortcut by using sophisticated algorithms to derive automatically analyzed metadata that intelligently guides how an image graded at one nit level (say 4000 nits) can be adjusted to be perceptually similar to viewers watching a 1000 nit display. Highlights and saturation that are too bright for a particular display will be adjusted to provide as close to the same experience without clipping or flattening image detail.

Furthermore, this automatic analysis can be manually trimmed by a colorist to account for the artistic intentions of the authors of a program, in cases where the automatic analysis doesn't do exactly what's wanted. This combination of auto-analysis and manual trimming is key to how Dolby Vision streamlines the process of mastering programs to accommodate backward compatibility with SDR displays, as well as the varying peak luminance capabilities of different makes and models of HDR consumer displays, both now and in the future. You're only required to make a 100 nit trim pass to guide the HDR program's conversion all the way down to SDR, and the Dolby Vision system can use that information to guide how intermediate presentations (such as at 700 or 1200 nits) should be adjusted. You can even do multiple trim passes at specific nit levels, such as a 100 nit pass and a 1000 nit pass, to give the Dolby Vision system more information to accurately guide intermediate presentations on different displays. Additionally, you don't have to trim every clip. If the analysis is good, you can skip those clips and only trim clips that need it. The overall system has been created to make it as efficient as possible for colorists to ensure that the widest variety of viewers see the image as it's meant to be seen.

This, in a nutshell, is the advantage of the Dolby Vision system. You can grade a program on a more future-proofed 4000 nit display, and use auto-analysis plus one or two manual trim passes to make the program backward compatible with SDR televisions, and capable of intelligently scaling the HDR highlights to provide the best representation of the mastered image for whatever peak luminance and color volume a particular television is capable of. All of this is guided by decisions made by the colorist during the grade.

At the time of this writing, all seven major Hollywood studios are mastering in Dolby Vision for cinema. Studios that have pledged support to master content in Dolby Vision for home distribution include Universal, Warner Brothers, Sony Pictures, and MGM. Content providers that have agreed to distribute streaming Dolby Vision content include Netflix, Vudu, and Amazon. If you want to watch Dolby Vision content on television at home, consumer television manufacturers LG, TCL, Vizio, HiSense, Sony, Toshiba, and Bang & Olfusen have all shipped models with Dolby Vision support.

Organizing Your Timeline for Dolby Vision Mastering

One of the first things you need to do before doing a Dolby Vision grade is to organize your timeline accordingly. Because each clip undergoes a visual analysis to facilitate the Dolby Vision workflow, there are specific limitations to how clips can appear in a timeline.

- All clips to be analyzed in a Dolby Vision workflow need to be on video track V1; clips on other tracks will be ignored.
- All clips that overlap one another as part of a composite must be turned into a single item in the timeline in order to be correctly analyzed. This means that each group of clips that create a composite in a timeline, be it multiple overlapping clips combined via keys or alpha channel transparency, multiple overlapping clips combined using composite or blend modes, or text generators appearing above one or more video clips, must be turned into a compound clip for Dolby Vision analysis to work correctly.

Letterboxing for Dolby Vision Mastering

The analysis of clips in a Dolby Vision workflow keeps track of the timeline aspect ratio, as well as the image aspect ratio of each clip in that timeline. Programs that mix different aspect ratios of letterboxing (or blanking) will be accommodated by the Dolby Vision analysis, however Dolby Vision does not support letterbox on two sides (both pillarbox and letterbox), only one at a time.

Setting Up Color Management for Dolby Vision Mastering

For an HDR signal to look correct, you need to output your graded program using the right EOTF for the HDR standard you're mastering. The EOTF maps the different levels DaVinci Resolve outputs to your HDR display using the SMPTE ST.2084 PQ setting required for outputting Dolby Vision. You can set this up in one of three different ways, as:

- Output Color Space and Gamma settings in RCM or ACES
- Color Space and Gamma settings within a series of Resolve FX Color Transform plugins that can be used at the end of each grade or at the end of a Timeline grade
- 3D LUTs used for converting signals from one standard to another that can be used at the end of each grade or at the end of a Timeline grade

While Dolby Vision content is not limited to a particular color space, Resolve Color Management provides a P3 D65 setting that matches the capabilities of most mastering displays in use at the time of this writing.

Choosing Mastering Displays for Dolby Vision

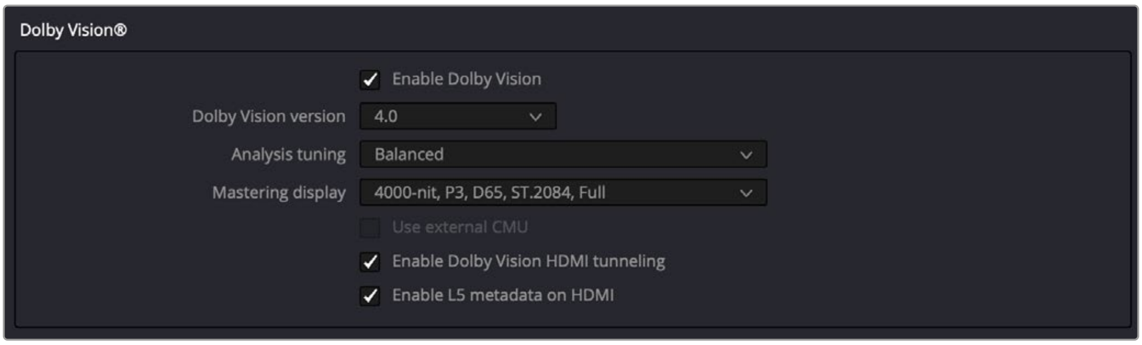
To do HDR grading, you need a suitable HDR display. Technically any monitor that supports SMPTE ST.2084 (aka PQ) will work. Happily, a growing number of professional displays from Sony, Flanders Scientific, TV-Logic, Canon, and Eizo are suitable for use in HDR grading suites. EBU Tech 3320 specifies the requirements for a Grade 1 HDR mastering monitor. Dolby recommends the following minimum requirements for HDR monitors:

- A minimum Peak Luminance of 1000 nits
- A 200,000:1 contrast ratio
- Minimum black at 0.005 nits
- Capable of at least 99% of P3 gamut

For more information on Dolby best practices for color grading Dolby Vision, visit: https://www.dolby.com/us/en/technologies/dolby-vision/dolby-vision-for-creative-professionals.html.

Using the Dolby Vision Internal Content Mapping Unit (iCMU)

DaVinci Resolve has a GPU-accelerated “internal” software version of the Dolby Vision CMU (Content Mapping Unit) for previewing Dolby Vision mapping right in DaVinci Resolve. iCMU support can be enabled and set up in the Color Management panel of the Project Settings by turning on the Enable Dolby Vision checkbox. This is a DaVinci Resolve Studio-only feature.



Dolby Vision settings in the Color Management panel of the Project Settings

The Dolby Vision group of settings also exposes menus for choosing the version of Dolby Vision you want to use, the type of analysis and what kind of Master Display you’re using, and whether or not to use an eCMU (assuming you possess the option). In addition, there are options to enable HDMI tunneling (as described below). Finally, turning Dolby Vision on also enables the Dolby Vision palette and controls in the Color page, which are described in greater detail later in this chapter.

To master with Dolby Vision in DaVinci Resolve using the built-in iCMU, you still need a more specific hardware setup than the average grading and finishing workstation, consisting of the following equipment:

- Your DaVinci Resolve grading workstation using a DeckLink 8K Pro or DeckLink 4K Extreme 12G video interface.
- A mastering display capable of outputting HDR nit levels suitable for the deliverable you’re required to produce

Simultaneous Master and Target Display Output for Dolby Vision

When mastering HDR and trimming versions for more limited displays, it's extremely useful to be able to evaluate your HDR grade and SDR trim pass side-by-side. It's possible to output both the Master Display output and the Target Display output simultaneously when you're grading with either Dolby Vision or HDR10+ enabled.

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Necessary Hardware

To work in this manner, you must have the following equipment:

- Your DaVinci Resolve grading workstation must output via a DeckLink 8K Pro via the attached HDMI card. This will pass the Dolby Vision metadata through to the SDR display, so you will see the SDR out from the HDR grade.
- Your Mastering Display must be capable of HDR nit levels suitable for the deliverable you're required to produce.
- A display that can be set to output calibrated SDR, probably using the BT.709 gamut

Enabling Simultaneous Monitoring

When you set up your display hardware, the HDR Master Display must be connected to output A, and the Target Display must be connected to output B of whichever BMD video output device you're using. Then, you need to turn on the "Use dual outputs on SDI" checkbox in the Master Settings of the Project Settings. At this point, assuming all of your connections are compatible with one another, you should see an HDR image output to your HDR display, and a trimmed image output to your SDR display.

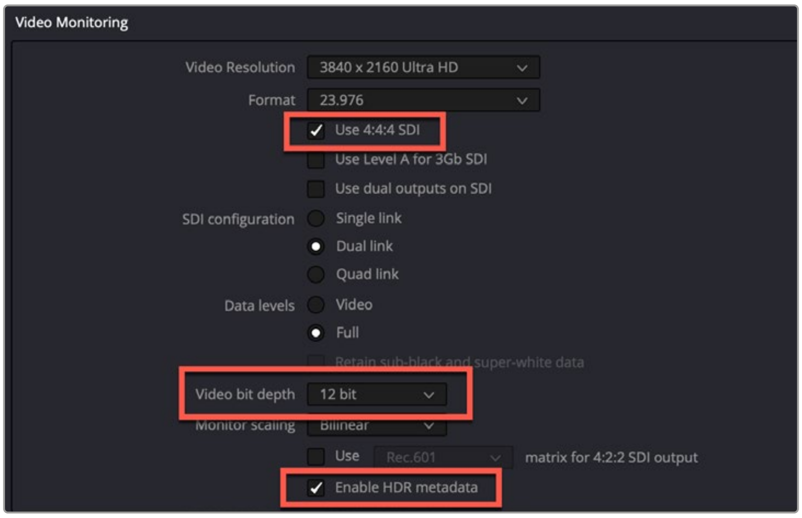
HDMI Tunneling using DeckLink 8K Pro G2

HDMI Tunneling allows you to preview Dolby Vision content directly on a consumer display in real time. The variables involved in display type and manufacturer all have their own processing behaviors, so HDMI tunneling is not for reference grading but more of letting you get an idea of what the audience will be seeing at home for QC purposes.

Performing HDMI tunneling in DaVinci Resolve is possible using lower cost Blackmagic DeckLink cards instead of Dolby's dedicated eCMU hardware. Currently, the only DeckLink card that supports Dolby Vision HDMI tunneling is the DeckLink 8K Pro G2.

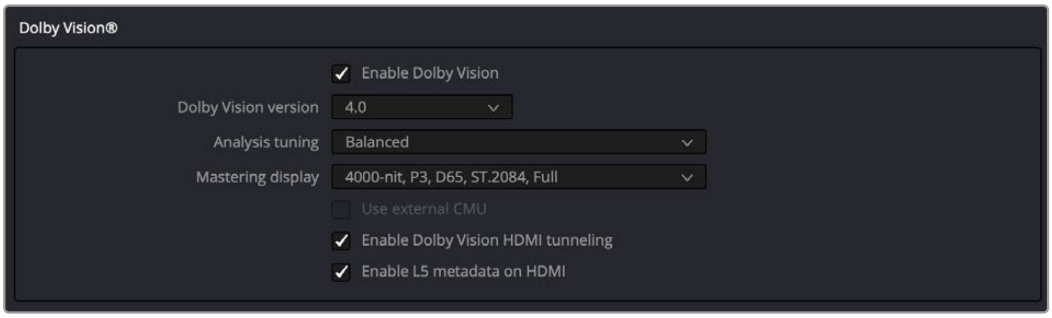
Setting up Dolby Vision HDMI Tunneling in DaVinci Resolve:

1. In Preferences > System > Video and Audio I/O, make sure that both the Capture Device and Monitor Device are set to the DeckLink 8K Pro.
2. In Project Settings > Master Settings > Video Monitoring, set the following parameters:
    - **Use 4:4:4 SDI:** Check this box.
    - **Video bit depth:** Set this value to 12 bit.
    - **Enable HDR metadata:** Check this box.



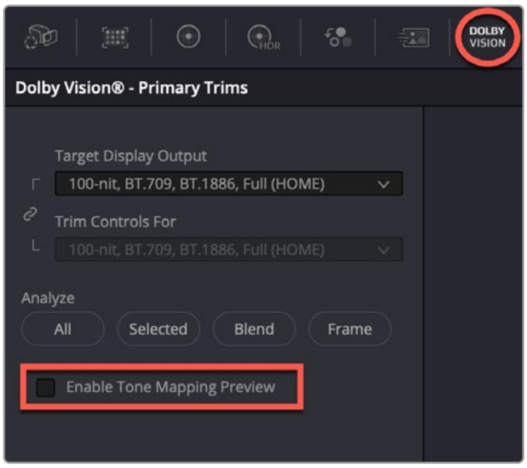
The Project settings in Video Monitoring to enable Dolby Vision HDMI tunneling

3. In Project Settings > Color Management > Dolby Vision, set the following parameters:
   - **Enable Dolby Vision:** Check this box.
   - **Enable Dolby Vision HDMI tunneling:** Check this box.
   - **Enable L5 metadata on HDMI:** Check this box.



The Project settings in Dolby Vision to enable HDMI tunneling

4. In the Color page, in the Dolby Vision palette, make sure that the Enable Tone Mapping Preview is disabled (the box is not checked). HDMI tunneling will not work if this box is checked.



Disable the Enable Tone Mapping Preview box
in the Dolby Vision palette in the Color page

For more information on Dolby Vision HDMI tunneling go to Dolby's website:
https://professionalsupport.dolby.com/s/article/HDMI-Tunneling?language=en_US

External Content Mapping Unit (eCMU) for Dolby Vision

DaVinci Resolve supports the use of a Dolby External Content Mapping Unit (eCMU) for studios doing more intensive HDR mastering work, as it lets you monitor and adjust an HDR display simultaneously to an SDR display for side-by-side trimming at high resolutions via hardware. The eCMU also has the ability to preview Dolby Vision on a consumer display in real time via HDMI tunneling to view directly what the audience will see at home.

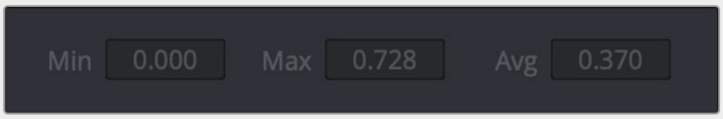
Auto Analysis is Available to All Studio Users

Resolve Studio enables either unlicensed or licensed users to automatically analyze the image and generate Dolby Vision analysis metadata. This metadata is used to deliver Dolby Vision content and to render other HDR and SDR deliverables from the HDR grade that you've made. This enables any DaVinci Resolve Studio user to create Dolby Vision deliverables with Level 1 metadata. However, manual trimming of the analysis metadata requires a license from Dolby.

The commands governing Dolby Vision auto-analysis, which are available to all Resolve Studio users, are available in the Color > Dolby Vision™ submenu, as well as the Dolby Vision palette, and consist of the following:

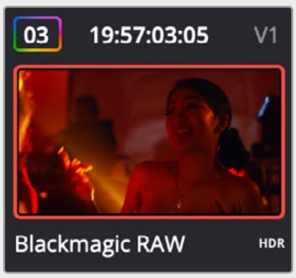
- **Analyze All Shots:** Automatically analyzes each clip in the Timeline and stores the results individually.
- **Analyze Selected Shot(s):** Only analyzes selected shots in the Timeline.
- **Analyze Selected And Blend:** Analyzes multiple selected shots as if they were a single sequence. The result is the same analysis being saved to each clip. Useful to save time when analyzing multiple clips that have identical content.
- **Analyze Current Frame:** A fast way to analyze clips where a single frame is representative of the entire shot.

Once you analyze a clip, the Min, Max, and Average fields automatically populate with the resulting L1 data; these fields are not editable.



The metadata fields for each clip

Additionally, clips that have been analyzed show an HDR badge in the Thumbnail timeline, to help you keep track of which clips have been analyzed and which have yet to be.



Analyzed clips have HDR badges to identify them

Licensing DaVinci Resolve to Expose Dolby Vision Trim Controls

To expose the Dolby Vision controls in DaVinci Resolve Studio that let you make manual trims on top of the automatic analysis that any copy of DaVinci Resolve Studio can do, you must email dolbyvisionmastering@dolby.com to receive more information about obtaining a license.

Once you've obtained a license file from Dolby, you can import it by choosing File > Dolby Vision > Load License, and its successful installation will enable the Dolby Vision controls to be enabled in the Color page. You should also receive a display configuration file, which can be loaded via the File > Dolby Vision > Load Configuration command and lets you populate the Dolby Vision drop-down menus with the most up to date options.

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Dolby Vision® Trim Controls in DaVinci Resolve

Once you've analyzed a clip, you're in a position to trim the result. The latest version of the Dolby Vision palette exposes four sets of controls. The first are the main controls:

- **Target Display Output:** This drop-down specifies what Dolby refers to as the Target Display, used to display the tone mapped image. This menu lets you choose specific display properties to obtain a preview of what the trimmed image will look like on different displays with different gamuts and peak luminance capabilities. You can link the Target Display Output and the Trim Controls For fields by clicking on the little chain icon directly to the left of the options. This automatically changes the Trim Controls For field to the same value you set in the Target Display Output, ensuring your trim controls are always right for the selected display.
- **Trim Controls for:** Which Target Display you're currently trimming for. The default setting (100-nit, BT.709, BT.1886, Full) lets you monitor an SDR version of the HDR image, so you can see how the trim metadata tone maps the image on non-HDR televisions.
- **Analyze controls:** The commands governing Dolby Vision auto-analysis are available as buttons, which perform the same functions as their similarly named counterparts in the Color > Dolby Vision submenu. Please note that most trim controls are disabled until you perform an analysis, which is a necessary first step.

  All: Automatically analyzes each clip in the current Timeline and stores the results individually.

  Selected: Only analyzes selected shots in the Timeline.

  Blend: Analyzes multiple selected shots as if they were a single sequence. The result is the same analysis being saved to each clip. You need to use the blend option when analyzing two clips that meet at a through edit separating otherwise contiguous frames. It's also typical to use the Blend option when analyzing a scene of clips that take place at the same location at the same time, to ensure that natural variations in lighting don't add unwanted variations between the analyses of clips that are supposed to already be balanced with one another. Blend also saves time when analyzing multiple clips that have identical content.

  Frame: Useful in situations where part of a clip has an extreme level of color or lightness that's not typical of the rest of the clip, that incorrectly biases the analysis and produces a poor result. Placing the playhead on a frame that's representative of how the clip is supposed to look and using the Frame option bases the analysis on only that frame. This is also a fast way to analyze clips where a single frame is representative of the entire shot.

- **Enable Tone Mapping Preview:** Lets you see the target display output in the Color page Viewer and video output, so you can evaluate how the tone mapped version looks on your HDR display. This control is disabled when you enable "Use dual-outputs on SDI" in the Master Settings of the Project Settings, since the second output SDI now automatically displays the target display output.
- **Mid Tone Offset (CM v4.0 only):** This control is used to match the overall exposure between the tone mapped SDR signal to the HDR master. This offset is applied to the L1 Mid values, allowing the adjustment of mid tones without affecting the blacks and highlights. It can be used to shift

overall L1 analysis to ensure the best preservation of artistic intent. This setting is shared among all trim passes you do at all nit levels, so if you've done two trim passes, one at 100 nits and another at 1000 nits, adjusting this setting always adjusts both trim passes at once. Changes made to this control are recorded to the L3 metadata for each clip.

The second are the Min, Mid, and Max metadata fields that are populated by the analyzed values of the current clip. These fields cannot be edited, although analysis metadata can be copied and pasted among clips. These values represent the L1 analysis and are used to calculate how the HDR image should be trimmed to fit into the video standard specified by the Target Display.

The third are the Primary Trims, which are only editable if you've performed an analysis and if you have a license from Dolby. Which controls are exposed depends on the version of Dolby Vision you've selected in the Color Management panel of the Project Settings.

MEDIA

Dolby Vision CM v2.9 Controls

If you choose Dolby Vision 2.9 in the Color Management panel of the Project Settings, it activates the 2.9 version of Dolby's content mapping algorithm and exposes the original Dolby Vision trim controls. It is no longer suggested to use these, since you can do a Dolby Vision 4.0 analysis and trim, and still export converted 2.9 metadata for legacy workflows.

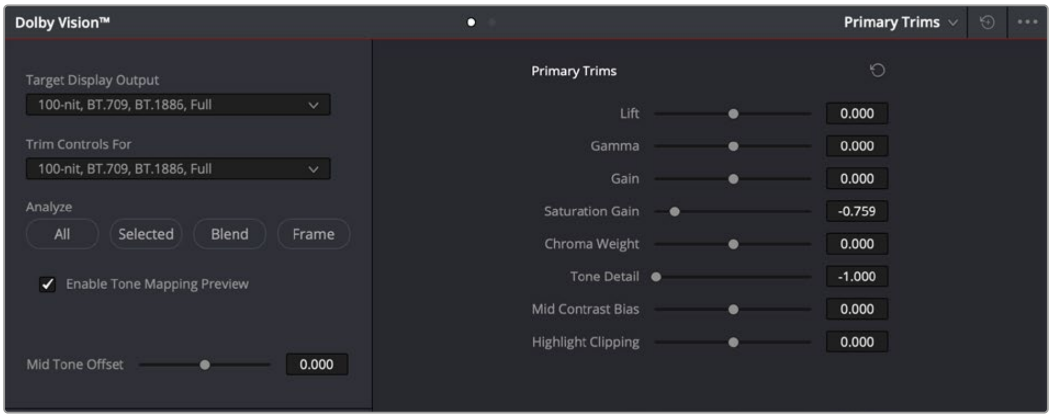
- **Lift/Gamma/Gain:** These controls function similarly to the Y-only Lift, Gamma, and Gain master wheels of the Color Wheels palette, to let you trim the overall contrast levels of the image. The Dolby Best Practices Guide recommends to limit positive Lift to no more than 0.025, and mostly restrict yourself to using Gamma and Gain if necessary to lighten the image.
- **Saturation Gain:** Lets you trim the saturation of the most highly saturated areas within a scene. Lesser saturated values will be less affected.
- **Chroma Weight:** Darkens saturated parts of the image to preserve colorfulness in areas of the image that are clipped by smaller gamuts that don't have enough headroom for saturation in the highlights.
- **Tone Detail:** Lets you preserve contrast detail in the highlights that might otherwise be lost when the highlights are mapped to lower dynamic ranges, usually due to clipping. Increasing Tone Detail Weight increases the amount of highlight detail that's preserved. When used, can have the effect of sharpening highlight detail.

Dolby Vision CM v4.0 Controls

If you choose Dolby Vision 4.0 in the Color Management panel of the Project Settings, it activates the 4.0 version of Dolby's content mapping algorithm, and exposes the following controls.

- **Lift/Gamma/Gain:** These controls function similarly to the Y-only Lift, Gamma, and Gain master wheels of the Color Wheels palette, to let you trim the overall contrast levels of the image. The Dolby Best Practices Guide recommends to limit positive Lift to no more than 0.025, and mostly restrict yourself to using Gamma and Gain if necessary to lighten the image.
- **Saturation Gain:** Lets you trim the saturation of the most highly saturated areas within a scene. Lesser saturated values will be less affected.
- **Chroma Weight:** Darkens saturated parts of the image to preserve colorfulness in areas of the image that are clipped by smaller gamuts that don't have enough headroom for saturation in the highlights.
- **Tone Detail:** Lets you preserve contrast detail in the highlights that might otherwise be lost when the highlights are mapped to lower dynamic ranges, usually due to clipping. Increasing Tone Detail Weight increases the amount of highlight detail that's preserved. When used, can have the effect of sharpening highlight detail.
- **Mid Contrast Bias:** Affects image contrast in the region around the computed average picture level. This lets you increase or decrease contrast in the midtones of the image.

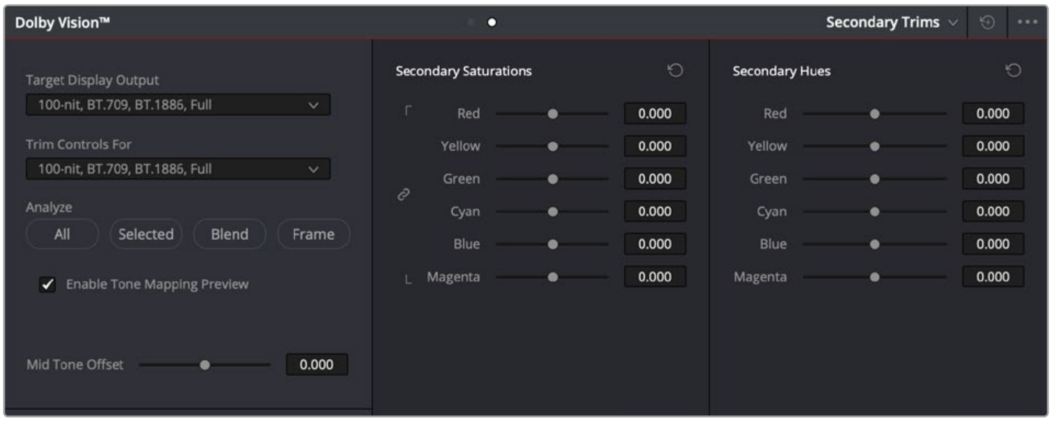
- **Highlight Clipping:** Reduces details and affects the roll-off the brighter part of the image by clipping the highlights as required. This is useful when the tone mapped image is displaying unwanted details.



The Primary Trims controls that are found in the Dolby Vision palette are only enabled once you've authorized your system with a special license, available from Dolby.

The fourth set of controls is available via a second palette mode, the Secondary Trims. These are only editable if you've performed an analysis and if you have a license from Dolby.

- **Secondary Saturations:** A set of slider-based vector-style controls (similar to the Hue vs. Sat curve) lets you adjust the Saturation of Red, Yellow, Green, Cyan, Blue, and Magenta to help you selectively fine tune the results.
- **Secondary Hues:** Another set of slider-based vector-style controls (similar to the Hue vs. Hue≈controls) lets you adjust the Hue of Red, Yellow, Green, Cyan, Blue, and Magenta to help you fine tune the results.



The Secondary Trims controls, as seen on a licensed Dolby Vision system

Together, all of this trimming metadata lets the colorist guide how the iCMU or eCMU transforms the image from the Mastering Display specified in the Project Settings to the Target Display specified in the Dolby Vision palette. This metadata is carried throughout the ecosystem so that your artistic intent is preserved on a variety of platforms and displays.

Previewing and Trimming At Different Levels

Additionally, the iCMU or eCMU can be used to preview 100 nit, 600 nit, 1000 nit, and 2000 nit versions of your program, with different gamuts, if you want to see how your master will scale to those combinations of peak luminance levels and standards. This, of course, requires your DaVinci Resolve workstation or eCMU to be connected to a display that's capable of being set to those peak luminance output levels.

Though it's not at all typical, you also have the option to set the "Trim Controls For" drop-down menu to different combinations of peak luminance, gamut, and color temperature, in order to visually trim the grades of your program at up to four different peak luminance levels, including 100 nit, 600 nit, 1000 nit, and 2000 nit reference points. Choosing a setting from the "Trim Controls For" drop-down menu sets you up to adjust trim metadata for that setting.

Choosing different settings from the "Trim Controls For" drop-down menu lets you can optimize a program's visuals for the peak luminance and color volume performance of many different televisions with a much finer degree of control. If you take this extra step of doing a complete trim pass of your program at multiple nit levels (using the Dolby Vision controls), the Level 2, or Level 8 metadata you generate in each trim pass ensures that the artistic intent is preserved as closely as possible across a wide variety of displays, in an attempt to provide the viewer with the best possible representation of the director's intent, no matter where it appears.

> For example, if a program were graded relative to a 4000 nit display, along with a single 100 nit BT.709 trim pass, then a Dolby Vision-compatible television with 750 nit peak output will reference the 100 nit trim pass metadata in order to come up with the best way of "splitting the difference" to output the signal correctly. On the other hand, were the colorist to do three trim passes, the first at 100 nits, -cond at 600 nits, and a third at 1000 nits, then a 750 nit-capable Dolby Vision television would be able to use the 600 and 1000 nit trim metadata to output more accurately scaled color volume and HDR-strength highlights, relative to the colorist's adjustments, that take better advantage of the 750 nit output of that television.

Managing Dolby Vision Metadata

As you go through the process of analyzing and trimming the HDR grades displayed on your Master Display to look appropriate on your Target Display, you'll sometimes find it useful to copy and paste metadata from one clip to another. You can copy and paste Analysis Metadata separately from Trim Metadata and Mid Tone Offset, and you can choose to copy and paste metadata for all Target Displays when you're trimming multiple passes, or you can copy and paste metadata for only the current Target Display if you're trimming multiple passes and you only want to overwrite metadata for a single pass.

Methods of Copying and Pasting Dolby Vision Metadata:

To copy and paste Analysis Metadata: Select a clip you want to copy from, choose Copy Analysis Metadata from the Dolby Vision palette option menu, then select a clip you want to paste to, and choose Paste Analysis Metadata from the Dolby Vision palette option menu.

To copy and paste Trim Metadata for all Target Displays: Do one of the following:

- Select a clip you want to copy from, choose Edit > Dolby Vision > Copy Trim Metadata, then select a clip you want to paste to, and choose Edit > Dolby Vision > Paste Trim Metadata.
- Select a clip you want to copy from, choose Copy Trim Metadata from the Dolby Vision palette option menu, then select a clip you want to paste to, and choose Paste Trim Metadata from the Dolby Vision palette option menu.
- Select a clip you want to paste to, then press and hold the Option-Shift keys, and middle-click the clip you want to copy from.

To copy and paste Trim Metadata for the current Target Display: Do one of the following:

- Select a clip you want to copy from, choose Copy Trim Metadata from the Dolby Vision palette option menu, then select a clip you want to paste to, and choose Paste Trim Metadata to Current from the Dolby Vision palette option menu.
- Select a clip you want to paste to, then press and hold the Option key, and middle-click the clip you want to copy from.

To copy and paste Mid Tone Offset: Select a clip you want to copy from, choose Copy Mid Tone Offset from the Dolby Vision palette option menu, then select a clip you want to paste to, and choose Paste Mid Tone Offset from the Dolby Vision palette option menu.

MEDIA

Setting Up Resolve Color Management for Grading HDR

Once the hardware is set up, setting up Resolve itself to output HDR for Dolby Vision mastering is easy using Resolve Color Management (RCM). This procedure is pretty much the same no matter which HDR mastering technology you're using; only specific Output Color Space settings will differ.

1. Set Color Science to DaVinci YRGB Color Managed in the Color Management panel of the Project Settings.
2. Then, open the Color Management panel, and set the Output Color Space drop-down to the ST.2084 setting that corresponds to the peak luminance, in nits, of the grading display you're using. For example, if you're grading with a Sony BVM X300, choose ST.2084 1000 nit, but if you're grading with a Flanders Scientific XM310K, choose ST.2084 3000 nit, in order to use the full capabilities of each display. Be aware that whichever HDR setting you choose will impose a hard clip at the maximum nit value supported by that setting. This is to prevent accidentally overdriving HDR displays for which there are negative consequences (not all HDR displays have this limitation).
   - ST.2084 300 nit
   - ST.2084 500 nit
   - ST.2084 800 nit
   - ST.2084 1000 nit
   - ST.2084 2000 nit
   - ST.2084 3000 nit
   - ST.2084 4000 nit

   This setting is only the output EOTF (a sort of gamma transform, if you will, using the terminology that DaVinci Resolve's UI has used up until now).
3. Next, choose a setting in the Timeline Color Space that corresponds to the gamut you want to use for grading, and that will be output. For example, if you want to grade the Timeline as a log-encoded signal and "normalize" it yourself, you can choose ARRI Log C or Cineon Film Log (this workflow is highly recommended for the best results). If you would rather save time by having DaVinci Resolve normalize the Timeline to P3-D65 and grade that way, you can choose that setting as well. In terms of defining the output gamut, the rule is that if "Use Separate Color Space and Gamma" is turned off, the Timeline Color Space setting will define your output gamut. If "Use Separate Color Space and Gamma" is turned on, then you can specify whatever gamut you want in the left Output Color Space drop-down menu, and choose the EOTF from the right drop-down menu (as described in step 2).

4. Be aware that, when it's being properly output, HDR ST.2084 signals appear very "log-like," in order to pack a wide dynamic range into the bandwidth of a standard video signal. It's the HDR display itself that "normalizes" this log-encoded image to look as it should. For this reason, the image you see in your Color page Viewer is going to appear flat and log-like, even though the image being displayed on your HDR reference display looks vivid and correct. If you're using a typical SDR computer display, and you want to make the image in the Color Page Viewer look "normalized" at the expense of clipping the HDR highlights (in the Viewer, not in the grade), you can use the 3D Color Viewer Lookup Table setting in the Color Management panel of the Project Settings to assign the appropriate ST.2084 setting with a peak nit level that corresponds to the HDR broadcast display you're outputting to.
5. Additionally, the "Timeline resolution" and "Pixel aspect ratio" (in the project settings) that your project is set to use is saved to the Dolby Vision metadata, so make sure your project is set to the final Timeline resolution and PAR before you begin grading.

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DaVinci Resolve Grading Workflow For Dolby Vision

Once the hardware and software is all set up, you're ready to begin grading HDR. The workflow is fairly straightforward.

1. First, grade the HDR image on your HDR Monitor to look as you want it to. Dolby recommends starting by setting the look of the HDR image, to set the overall intention for the grade.
2. When using various grading controls in the Color page to grade HDR images, you may find it useful to enable the HDR Mode of the node you're working on by right-clicking that node in the Node Editor and choosing HDR Mode from the contextual menu. This setting adapts that node's controls to work within an expanded HDR range. Practically speaking, this makes controls that operate by letting you make adjustments at different tonal ranges, such as Custom Curves, Soft Clip, and so on, work more easily with wide-latitude signals.
3. When you're happy with the HDR grade, click the Analysis button in the Dolby Vision palette. This analyzes every pixel of every frame of the current shot, and performs and stores a statistical analysis that is sent to the iCMU or eCMU to guide its automatic conversion of the HDR signal to an SDR signal.
4. Choose "Target Display Output" and "Trim Controls For" settings that you want to trim to. By default, these are set to "100-nit, BT.709, BT.1886, Full," which is a typical SDR deliverable. However, other options are available if you want to do multiple trim passes to obtain a more accurate result. Whichever setting you choose from, "Trim Controls For" dictates which trim pass you're doing. You can do multiple trim passes by choosing another option from this menu.
5. If you're not happy with the automatic conversion, use the trim controls in the Dolby Vision palette to manually trim the result to the best possible BT.709 approximation of the HDR grade you created in step 1.
6. If you obtain a good result, then move on to the next shot and continue work. If you cannot obtain a good result, and worry that you may have gone too far with your HDR grade to derive an acceptable SDR tone mapping, you can always trim the HDR grade a bit, and then retrim the SDR grade to try and achieve a better tone mapping. Dolby recommends that if you make significant changes to the HDR master, particularly if you modify the blacks or the peak highlights, you should reanalyze the scene. However, if you only make small changes, then reanalyzing is not strictly required.

As you can see, the general idea promoted by Dolby is that a colorist will focus on grading the HDR picture relative to the 1000, 2000, 4000, or higher nit display that is being used, and will then rely on the colorist to use the Dolby Vision controls to "trim" this into a 100 nit SDR version. This metadata is saved as part of the mastered media, and it's used to more intelligently tone map the entire image to fit within any given display's parameters. The colorist's artistic intent is used to guide all dynamic adjustments to the content.

Delivering Dolby Vision

Once you're finished grading the HDR and trimming the SDR tone mapping, you need to output your program correctly in the Deliver page.

Rendering a Dolby Vision Master

To deliver a Dolby Vision master after you've finished grading, you want make sure that the Output Color Space of the Color Management panel of the Project Settings is set to the appropriate HDR ST.2084 setting based on the peak output you want to deliver (any values above will be clipped). Then, you want to set your render up to use one of the following Format/Codec combinations:

- TIFF, RGB 16-bit
- EXR, RBG-half (no compression)

When you render for tapeless delivery, all Dolby Vision metadata is recorded into a Dolby Vision XML and delivered along side either the Tiffs or EXR renders. To export a Dolby Vision XML file, select your timeline in the media pool and choose File > Export >Timeline. Navigate to where you want to save the file and select Dolby Vision v2.9 (or v4.0) MXF files from the file type selector and click save. These two sets of files are then delivered to a facility that's capable of creating the Dolby Vision deliverable file.

Rendering a Dolby Vision IMF

You can deliver directly to an IMF that includes an MXF with embedded Dolby Vision metadata in the package. To export a Dolby Vision IMF use the following Video settings in the Deliver page:

- **Format:** IMF
- **Codec:** Kakadu JPEG 2000
- **Type:** Dolby Vision (HD, 2K, UHD, or 4K) depending on your deliverable resolution.

  Configure the rest of the IMF settings as necessary for your project.

The Video settings to use for creating a Dolby Vision IMF in the Deliver page

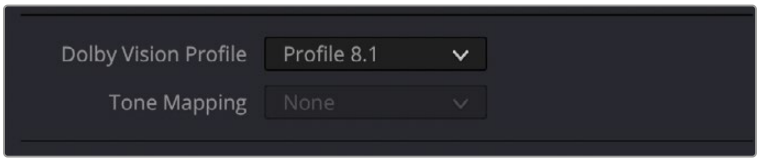
Rendering a Dolby Vision H.265 file

You can deliver directly to a Dolby Vision compatible H.265 file, which allows you to playback video that triggers playback in the Dolby Vision mode on your television or computer screen. To export a Dolby Vision H.265 file, use the following Video settings in the Deliver page:

- **Format:** MP4 or QuickTime
- **Codec:** H.265
- **Dolby Vision Profile:** Set the Dolby Vision profile you wish to use, or None to select the Tone Mapping manually.

- **Tone Mapping:** Choose None for no tone mapping, or Dolby Vision to expose a list of common deliverables to tone map to.

  Configure the rest of the H.265 settings as necessary for your project.

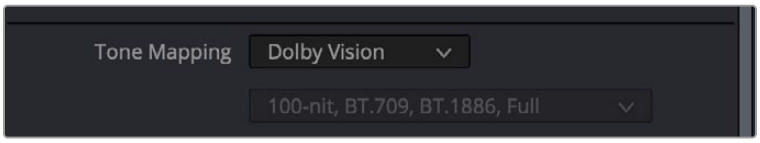


The Dolby Vision Profile and Tone Mapping settings in the H.265 Video section of the Deliver page

Rendering an Ordinary SDR Media File or Other Specific HDR Trim Pass

If you want to export the SDR trim pass, then you can choose Dolby Vision from the Tone Mapping drop-down menu in the Advanced Settings of the Render Settings list on the Deliver page, and choose the 100-nit, BT.709, BT.1886, Full setting below. With this enabled, you can output the SDR version of your program to any format you like.

You can also export the trims for other HDR nit levels for specific displays, at 600, 1000 or 2000 nits and in the either the BT.2020 or P3 gamuts.



The Tone Mapping setting in the Advanced Settings of the Render Settings list

SMPTE ST.2084 and HDR10

Many display manufacturers who have no interest in licensing Dolby Vision for inclusion in their displays are instead going with the simpler method of engineering their displays to be compatible with SMPTE ST.2084. It requires only a single stream for distribution, there are no licensing fees, no special hardware is required to master for it (other than an HDR mastering display), and there's no special metadata to write or deal with.

Interestingly, SMPTE ST.2084 ratifies the "PQ" EOTF that was originally developed by Dolby, and which is used by Dolby Vision, into a general standard that accommodates encoding HDR at peak luminance values up to 10,000 cd/m^2. This standard requires at minimum a 10-bit signal for distribution, and the EOTF is mathematically described such that the video signal utilizes the available code values of a 10-bit signal as efficiently as possible, while allowing for such a wide range of luminance in the image.

SMPTE ST.2084 is also part of the "Ultra HD Premium" industry specification, which stipulates that televisions bearing the Ultra HD Premium logo have the following capabilities:

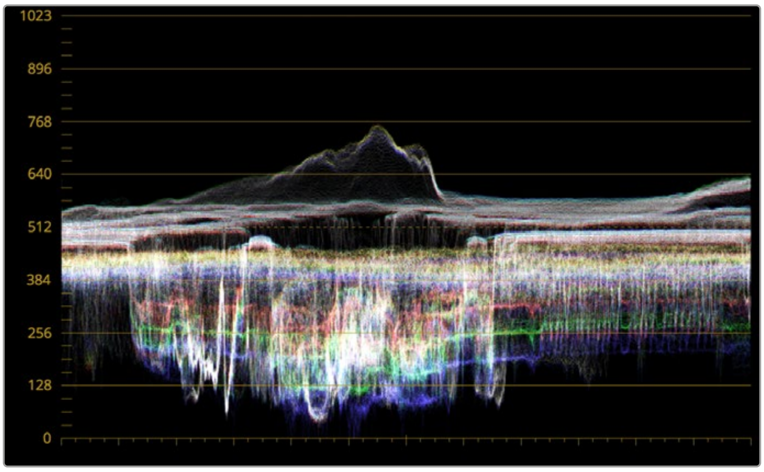
- A minimum UHD resolution of 3840 x 2160
- A minimum gamut of 90% of P3
- A minimum dynamic range of either 0.05 nits black to 1000 nits peak luminance (to accommodate LCD displays), or 0.0005 nits black to 540 nits peak luminance (to accommodate OLED displays)
- Compatibility with SMPTE ST.2084

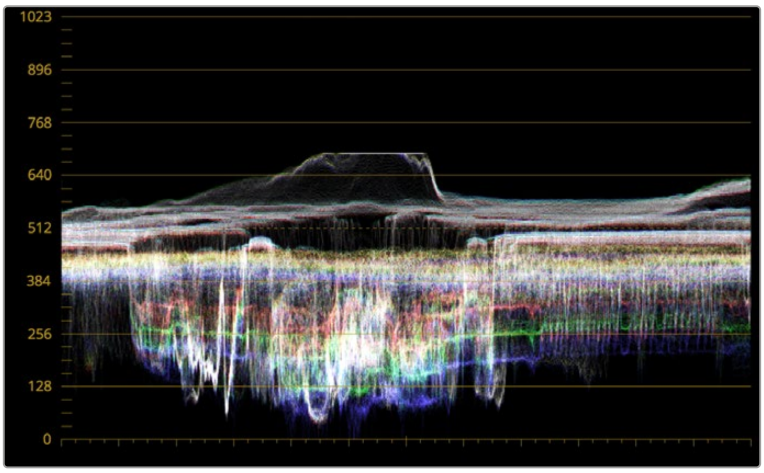
Finally, ST.2084 has been included in the HDR 10 standard adopted by the Blu-ray Disc Association (BDA) that covers Ultra HD Blu-ray. HDR 10 stipulates that Ultra HD Blu-ray discs have the following characteristics:

- UHD resolution of 3840 x 2160
- Up to the Rec. 2020 gamut
- SMPTE ST.2084
- Mastered with a peak luminance of 1000 nits

The downside is that, by itself, an HDR10 mastered program is not backward compatible with BT.709 displays using BT.1886 (although the emerging HDR10+ standard described later addresses this). Furthermore, no provision is made to scale the above-100 nit portion of the image to accommodate different displays with differing peak luminance levels. For example, if you grade and master an image to have peak luminance of 4000 nits, and you play that signal on an HDR10-compatible television (using ST.2084) that's only capable of 800 nits, then everything above 800 nits will be clipped, while everything below 800 nits will look exactly as it should relative to your grade.

This is because ST.2084 is referenced to absolute luminance. If you grade an HDR image referencing a 1000 nit peak luminance display, as is recommended by HDR10, then any display using ST.2084 will respect and reproduce all levels from the HDR signal that it's capable of reproducing as you graded them, up to the maximum peak luminance level it can reproduce. For example, on an HDR10-compatible television capable of outputting 500 nits, all mastered levels from 501–1000 will be clipped, as seen in the screenshot below.





Comparing the original 1000 nit waveform representing the grading monitor to a 500 nit clipped waveform representing the consumer television

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How much of a problem this is really depends on how you choose to grade your HDR-strength highlights. If you're only raising the most extreme peak highlights to maximum HDR-strength levels, then it's entirely possible that the audience might not notice that the display is only outputting 800 nits worth of signal and clipping any image details from 801–1000 nits because there weren't that many details above 800 anyway. Or, if you're grading large explosive fireballs up above 800 nits in their entirety because it looks cool, then maybe the audience will notice. The bottom line is, when you're grading for displays that are only capable of ST.2084, you need to think about these sorts of things.

Monitoring and Grading to ST.2084 in DaVinci Resolve

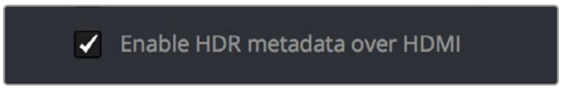
Monitoring an ST.2084 image is as simple as obtaining a ST.2084-compatible HDR display and connecting it to the output of your DeckLink 8K, DeckLink 4K Extreme 12G, or UltraStudio 4K Extreme.

Setting up Resolve Color Management to grade for ST.2084 is identical to setting up to grade for Dolby Vision. You'll also monitor the video scopes identically, and output a master identically, given that both standards rely upon the same PQ curve.

TIP: If you're monitoring with the built-in video scopes in DaVInci Resolve, you can turn on "HDR (ST.2084/HLG)" in the Waveform Scale Style settings in the Scopes option menu, which will replace the 10-bit scale of the video scopes with a scale based on nit values (cd/m^2) instead.

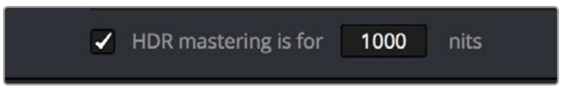
Connecting to HDR-Capable Displays using HDMI 2.0a

If you have a DeckLink 4K Extreme 12G or an UltraStudio 4K Extreme video interface, then DaVinci Resolve 12.5 and above can output the metadata necessary to correctly display HDR video signals to display devices using HDMI 2.0a when you turn on the "Enable HDR metadata over HDMI" checkbox in the Master Settings panel of the Project Settings.



The Enable HDR metadata over HDMI option in the Master Settings panel of the Project Settings lets you output HDR via HDMI 2.0a

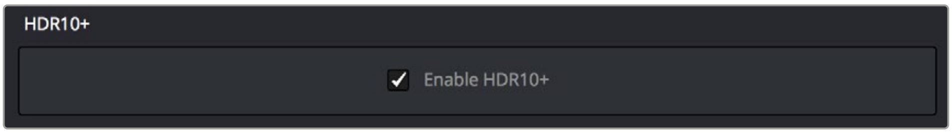
When you do so, a setting in the Color Management panel of the Project Settings, "HDR mastering is for X" lets you specify the output, in nits, to be inserted as metadata into the HDMI stream being output, so that the display you're connecting to correctly interprets it. The output you specify should match what your display is expecting.



The "HDR mastering is for" setting lets you insert metadata for HDR output via HDMI 2.0a

HDR10+™

DaVinci Resolve supports the new HDR10+ HDR format by Samsung. Please note that this support is a work in progress as this is a new standard. When enabled, an HDR10+ palette shows the results of the trimming analysis that make an automated downconversion of HDR to SDR, creating metadata to control how HDR-strength highlights look on a variety of supported televisions and displays. This is enabled and set up in the Color Management panel of the Project Settings with the Enable HDR10+ checkbox. Turning HDR10+ on enables the HDR 10+ palette in the Color page.



HDR 10+ settings in the Color Management panel of the Project Settings

Monitoring and Grading to ST.2084 for HDR10+

When you're grading a program for HDR10+ output, you'll need to monitor an ST.2084 image, which is as simple as obtaining a ST.2084-compatible HDR display and connecting it to the output of your DeckLink 8K, DeckLink 4K Extreme 12G, or UltraStudio 4K Extreme.

Setting up Resolve Color Management to grade for ST.2084 is identical to setting up to grade for Dolby Vision or regular HDR10. You'll also monitor the video scopes identically, and output a master identically, given that each of these standards rely upon the same PQ curve.

TIP: If you're monitoring with the built-in video scopes in DaVinci Resolve, you can turn on "HDR (ST.2084/HLG)" in the Waveform Scale Style settings in the Scopes option menu, which will replace the 10-bit scale of the video scopes with a scale based on nit values (cd/m^2) instead.

HDR10+ Grading Workflow

The idea behind the HDR10+ workflow is that you'll grade the HDR version of each clip in your program first, and then use the automatic analysis to create a downconverted tone mapped version of each shot that's controlled by metadata. Once the HDR10+ trim pass is complete, you'll deliver the rendered HDR output along with a set of HDR10+ JSON metadata files to a facility for final mastering.

Simultaneous Master and Target Display Output for HDR10+

When mastering HDR and trimming versions for more limited displays, it's extremely useful to be able to evaluate your HDR grade and tone mapped trim pass side by side. Starting in DaVinci Resolve 15, it's possible to output both the Master Display output and the Target Display output simultaneously when you're grading with either Dolby Vision or HDR10+ enabled.

Necessary Hardware

To work in this manner, you must have the following equipment:

- Your DaVinci Resolve grading workstation must output via a DeckLink 8K, DeckLink 4K Extreme 12G, UltraStudio 4K Extreme video interface, or better.
- Your Mastering Display must be capable of HDR nit levels suitable for the deliverable you're required to produce.
- An HDR target display that can be set to the appropriate tone mapped output.

Enabling Simultaneous Monitoring

When you set up your display hardware, the HDR Master Display must be connected to output A, and the Target Display must be connected to output B of whichever BMD video output device you're using. Then, you need to turn on the "Use dual outputs on SDI" checkbox in the Master Settings of the Project Settings. At this point, assuming all of your connections are compatible with one another, you should see an HDR image output to your HDR display and a trimmed image output to your SDR display.

HDR10+ Auto Analysis Commands

After you've graded an HDR version of each clip in your program, a set of HDR10+ specific commands let you auto-analyze each clip to create custom HDR to SDR downconversion metadata that give you a starting point for the SDR trim pass you need to do. These commands are available in the Color > HDR10+ submenu:

Analyze All Shots: Automatically analyzes each clip in the Timeline and stores the results individually.

Analyze Selected Shot(s): Only analyzes selected shots in the Timeline.

Analyze Selected and Blend: Analyzed multiple selected shots and averages the result, which is saved to each clip. Useful to save time when analyzing multiple clips that have identical content.

Analyze Current Frame: A fast way to analyze clips where a single frame is representative of the entire shot.

Delivering HDR10+

Once you're finished grading the HDR and trimming the SDR downconversion, you need to output your program correctly in the Deliver page.

Rendering an HDR10+ Master

To deliver an HDR10+ master after you've finished grading, you want make sure that the Output Color Space of the Color Management panel of the Project Settings is set to the appropriate HDR ST.2084 setting based on the peak output you want to deliver (any values above will be clipped). Then, you want to set your render up to use the highest quality Format/Codec combination that can be delivered to whomever is doing the final mastering.

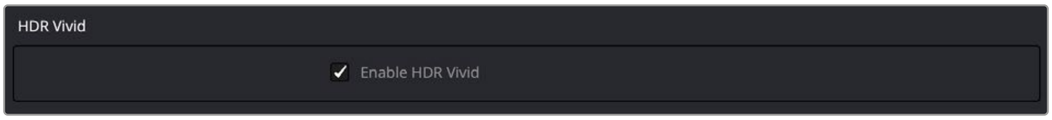
The HDR10+ analysis and manual trim metadata you generated while trimming is saved per clip, in a series of JSON sidecar files, which should then be exported by right-clicking that timeline in the Media Pool, and choosing Timelines > Export > HDR10+JSON.

These two sets of files are then delivered to a facility that's capable of creating an HDR10+ Mezzanine File (this cannot be done in DaVinci Resolve).

NOTE: The HDR10+ mastering workflow is still a work in progress. More information will be provided as it becomes available.

HDR Vivid

HDR Vivid is the HDR video technology standard released by the UHD World Association (UWA). Mastering in this format assures wide compatibility with HDR televisions, phones, computers, and other devices adhering to this standard.



HDR Vivid settings in the Color Management panel of the Project Settings

Monitoring and Grading to ST.2084 for HDR Vivid

When you're grading a program for HD Vivid output, you'll need to monitor an ST.2084 image, which is as simple as obtaining a ST.2084-compatible HDR display and connecting it to the output of your DeckLink 8K, DeckLink 4K Extreme 12G, or UltraStudio 4K Extreme.

Setting up Resolve Color Management to grade for ST.2084 is identical to setting up to grade for Dolby Vision or regular HDR10. You'll also monitor the video scopes identically, and output a master identically, given that each of these standards rely upon the same PQ curve.

TIP: If you're monitoring with the built-in video scopes in DaVinci Resolve, you can turn on "HDR (ST.2084/HLG)" in the Waveform Scale Style settings in the Scopes option menu, which will replace the 10-bit scale of the video scopes with a scale based on nit values (cd/m^2) instead.

HDR Vivid Grading Workflow

The idea behind the HDR Vivid workflow is that you'll grade the HDR version of each clip in your program first, and then use the automatic analysis to create a downconverted tone mapped version of each shot that's controlled by metadata. Once the HDR Vivid trim pass is complete, you'll deliver the rendered HDR output with embedded HDR Vivid metadata.

Simultaneous Master and Target Display Output for HDR Vivid

When mastering HDR and trimming versions for more limited displays, it's extremely useful to be able to evaluate your HDR grade and tone mapped trim pass side by side. Starting in DaVinci Resolve 15, it's possible to output both the Master Display output and the Target Display output simultaneously when you're grading with HDR Vivid enabled.

Necessary Hardware

To work in this manner, you must have the following equipment:

- Your DaVinci Resolve grading workstation must output via a DeckLink 8K, DeckLink 4K Extreme 12G, UltraStudio 4K Extreme video interface, or better.
- Your Mastering Display must be capable of HDR nit levels suitable for the deliverable you're required to produce.
- An HDR target display that can be set to the appropriate tone mapped output.

Enabling Simultaneous Monitoring

When you set up your display hardware, the HDR Master Display must be connected to output A, and the Target Display must be connected to output B of whichever BMD video output device you're using. Then, you need to turn on the "Use dual outputs on SDI" checkbox in the Master Settings of the Project Settings. At this point, assuming all of your connections are compatible with one another, you should see an HDR image output to your HDR display and a trimmed image output to your SDR display.

HDR Vivid Trim Controls in DaVinci Resolve

The latest version of the HDR Vivid palette exposes three sets of controls. The first are the main controls:

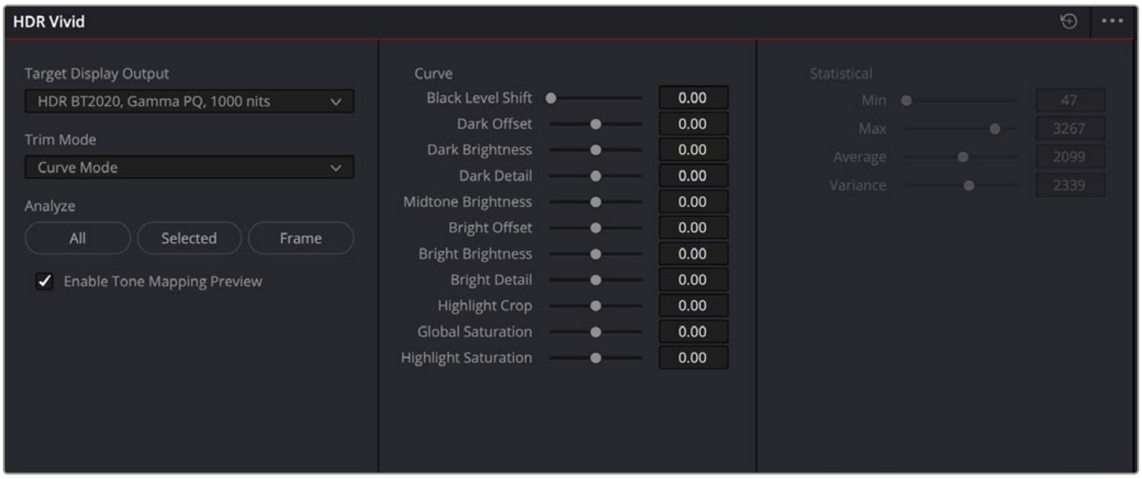
- **Target Display Output:** This drop-down specifies what parameters are used to display the tone mapped image. This menu lets you choose specific display properties to obtain a preview of what the trimmed image will look like on different displays with different gamuts and peak luminance capabilities.
- **Trim Mode:** Determines the toolset used to create the trims, either Curve or Statistical modes.
- **Analyze controls:** The commands governing HDR Vivid auto-analysis are available as buttons, which perform the same functions as their similarly named counterparts in the Color > HDR Vivid submenu. Please note that most trim controls are disabled until you perform an analysis, which is a necessary first step.

  All: Automatically analyzes each clip in the current Timeline and stores the results individually.

  Selected: Only analyzes selected shots in the Timeline.

  Frame: Useful in situations where part of a clip has an extreme level of color or lightness that's not typical of the rest of the clip, that incorrectly biases the analysis and produces a poor result. Placing the playhead on a frame that's representative of how the clip is supposed to look, and using the Frame option, bases the analysis on only that frame. This is also a fast way to analyze clips where a single frame is representative of the entire shot.
- **Enable Tone Mapping Preview:** Lets you see the target display output in the Color page Viewer and video output, so you can evaluate how the tone mapped version looks on your HDR display.
- The next set of controls activate based on which Trim Mode you have selected.
- **Curve:** The Curve trim mode exposes a variety of controls that lets the colorist adjust the trim metadata manually at the clip or frame level. Offset, Brightness, and Detail settings are available for the Dark and Bright parts of the curve. Midtones have a brightness control. Highlight Crop lets you bring any highlights back into range that may have exceeded the display's maximum brightness, thus blowing out. Separate Global and Highlight saturation settings are also possible.
- **Statistical:** The Statistical trim mode exposes controls that let you fine tune the automatic tone mapping algorithm used in the Analyze step.

MEDIA



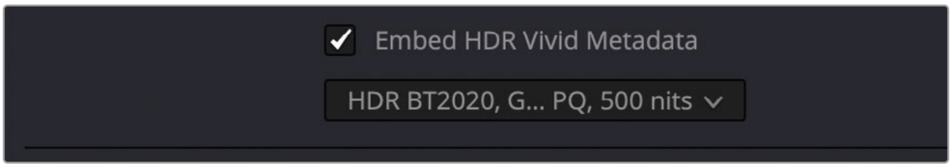
HDR Vivid controls in the Color page

Delivering HDR Vivid

Once you're finished grading the HDR and trimming the SDR downconversion, you need to output your program correctly in the Deliver page.

Rendering an HDR Vivid Master

To deliver an HDR Vivid master after you've finished grading, you want make sure that the Output Color Space of the Color Management panel of the Project Settings is set to the appropriate HDR ST.2084 setting based on the peak output you want to deliver (any values above will be clipped). Then, you want to set your render up to export an H.265 codec in the Video Settings, check the Embed HDR Vivid Metadata box, and select the appropriate HDR mode that you used to grade the master in the drop-down box below.



HDR Vivid settings in the Video section of the Deliver page

Hybrid Log-Gamma (HLG)

The BBC and NHK jointly developed another method of encoding HDR video, referred to as Hybrid Log-Gamma (HLG). The goal of HLG was to develop a method of mastering HDR video that would support a range of displays of different peak luminance capabilities without additional metadata, that could be broadcast via a single stream of data, that would fit into a 10-bit signal, and that in the words of the ITU-R Draft Recommendation BT.HDR, "offers a degree of compatibility with legacy displays by more closely matching the previous established television transfer curves."

The basic idea is that the HLG EOTF functions very similarly to BT.1886 from 0 to 0.6 of the signal (with a typical 0–1 range), while 0.6 to 1.0 smoothly segues into logarithmic encoding for the highlights. This means that, if you just send an HDR Hybrid Log-Gamma signal to an SDR display, you'd be able to see much of the image identically to the way it would appear on an HDR display, and the highlights would be compressed to present an acceptable amount of detail for SDR broadcast.

On a Hybrid Log-Gamma compatible HDR display, however, the log-like highlights of the image (not the BT.1886-like bottom portion of the signal, just the highlights) would be stretched back out, relative to whatever peak luminance level a given HDR television is capable of outputting, to return the image to its true HDR glory. This is different from the HDR10 method of distribution described previously, in which the graded signal is referenced to absolute luminance levels dictated by ST.2084, and levels that cannot be represented by a given display will be clipped.

And while this facility to support multiple HDR displays with differing peak luminance levels is somewhat analogous to Dolby Vision's ability to tailor HDR output to the unique peak luminance levels of any given Dolby Vision-compatible television, HLG requires no additional metadata to guide how the highlights are scaled, which depending on your point of view is either a benefit (less work), or a deficiency (no artistic guidance to make sure the highlights are being scaled in the best possible way).

As is true for most things, you don't get something for nothing. The BBC White Paper WHP 309 states that, for a 2000 cd/m^2 HDR display with a black level of 0.01 cd/m^2, up to 17.6 stops of dynamic range without visible quantization artifacts ("banding") is possible. BBC White Paper WHP 286 states that the proposed HLG EOTF should support displays up to about 5000 nits. So, partially, the backward compatibility that HLG makes possible is due in part to discarding long-term support for 10,000 nit displays. However, it's an open question whether or not over 5000 nits is even necessary for consumer enjoyment.

Sony, LG, Panasonic, JVC, Phillips, Hisense, Hitachi, and Toshiba have all either announced or are shipping consumer HDR televisions capable of displaying HLG encoded video, and of course DaVinci Resolve supports this standard through Resolve Color Management.

Grading Hybrid Log-Gamma in DaVinci Resolve

Monitoring an ST.2084 image is as simple as getting a Hybrid Log-Gamma-compatible HDR display, and connecting the output of your video interface to the input of the display.

Setting up Resolve Color Management to grade for HLG is identical to setting up to grade for Dolby Vision, except that there are four HLG settings to choose from for the Output Color Space:

- Rec.709 HLG ARIB STD-B67
- Rec.2020 HLG ARIB STD-B67
- Rec.2100 HLG
- Rec.2100 HLG (Scene)

Optionally, if you choose to enable "Use Separate Color Space and Gamma," you can choose either Rec. 2020 or Rec. 709 as your gamut, and Rec. 2100 HLG as your EOTF.

The levels you'll be monitoring in your scopes will be different from the table of data to nit values listed previously for grading to the PQ EOTF.

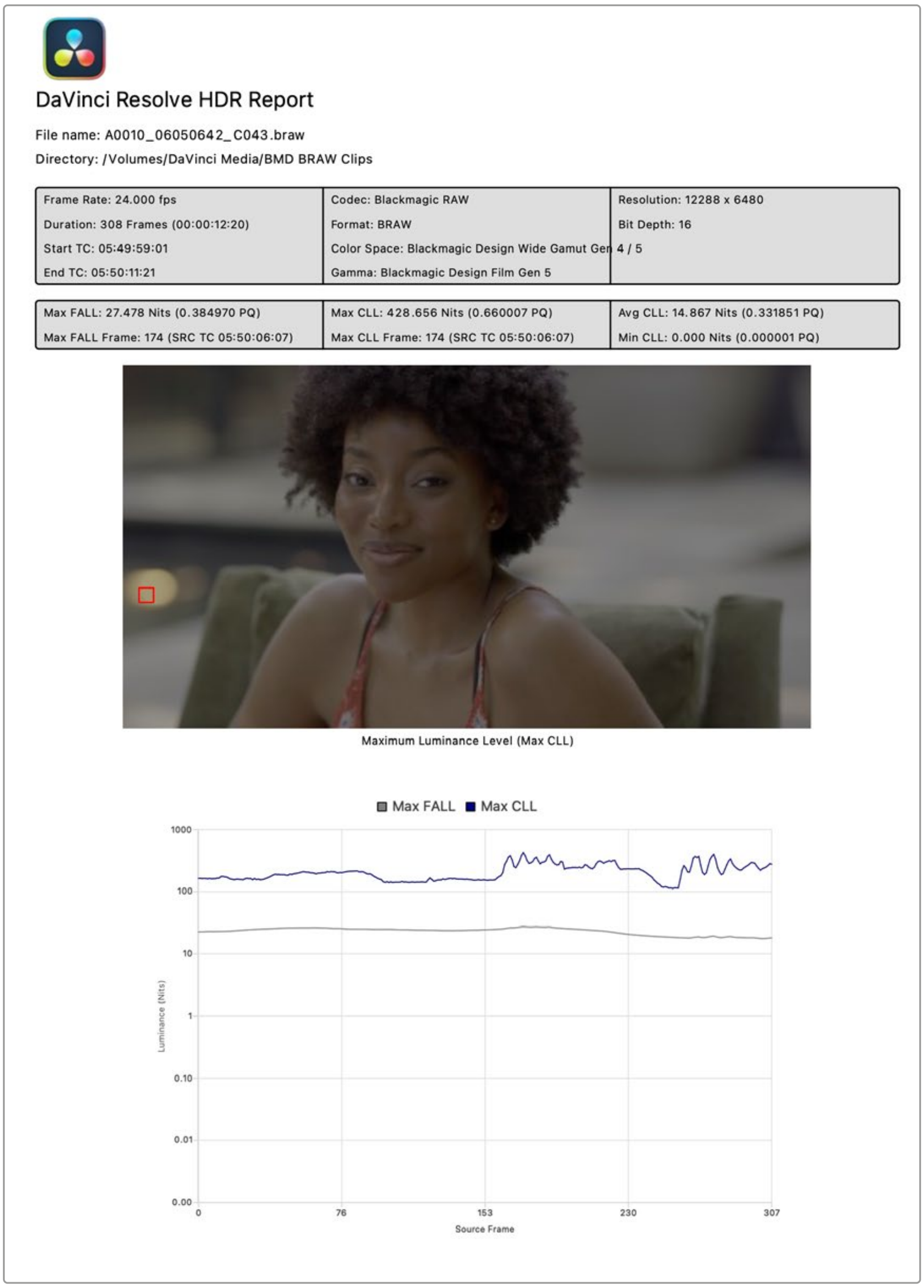
Ouputting Hybrid Log-Gamma

Once you've created an HLG grade for your program, you can output it to any high-quality 10-bit capable media format.

Generate HDR Light Level Report (Studio Version Only)

You can now generate HDR Light Level reports from the Media Pool. To do so, select a clip in the Media Pool, right-click it and choose Generate HDR Report from the context menu. The tool will prompt you to select a location to save the HDR Light Level report in .pdf format.

This tool analyzes all frames in rendered HDR clips and presents light level statistics, including MaxCLL and MaxFALL. It also includes a snapshot of the frame containing the maximum luminance level with a box highlighting the point where the maximum values were found.



DaVinci Resolve HDR Report

File name: A0010_06050642_C043.braw

Directory: /Volumes/DaVinci Media/BMD BRAW Clips

Frame Rate: 24.000 fps	Codec: Blackmagic RAW	Resolution: 12288 x 6480
Duration: 308 Frames (00:00:12:20)	Format: BRAW	Bit Depth: 16
Start TC: 05:49:59:01	Color Space: Blackmagic Design Wide Gamut Gen 4 / 5	
End TC: 05:50:11:21	Gamma: Blackmagic Design Film Gen 5	

Max FALL: 27.478 Nits (0.384970 PQ)	Max CLL: 428.656 Nits (0.660007 PQ)	Avg CLL: 14.867 Nits (0.331851 PQ)
Max FALL Frame: 174 (SRC TC 05:50:06:07)	Max CLL Frame: 174 (SRC TC 05:50:06:07)	Min CLL: 0.000 Nits (0.000001 PQ)

Maximum Luminance Level (Max CLL)

An HDR Light Level report exported from DaVinci Resolve